Paper Planes

How to fold a dart, and why it stays up

A sheet of paper is already a wing. Crease it well, throw it gently, and it will glide farther than it has any right to. This note is three pages: a fold, a reason, and nothing else.

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Turn the page. On a phone, swipe. Find the word "glide". Hold on a line and a friend who opened your invite sees a pointer on the same sentence.

Folding a dart

Letter or A4. Work on a table. Sharp creases fly straighter.

1. Fold the sheet in half the long way. Open it again. The centre crease is the spine.
2. Fold the top two corners down to the spine so they meet in a point. This is the nose.
3. Fold the long edges in to the spine. The nose gets sharper. Crease firmly with a thumbnail.
4. Fold the plane in half along the spine, printed side out.
5. Fold each wing down so the top edge of the wing lines up with the bottom of the fuselage. Leave a finger of body under the wings.
6. Pinch the fuselage. Hold it under the wings, a little behind the nose. Throw level, not up. A gentle toss.

Why it glides

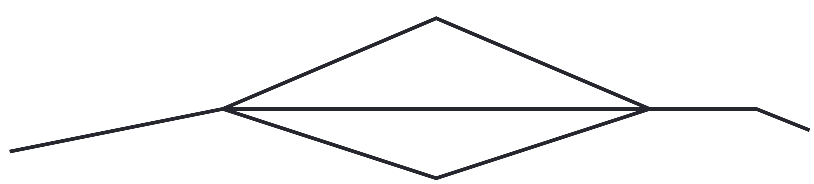
Four forces, same as any aeroplane, just quieter.

Lift comes from air moving faster over the top of the wing than under it. A paper dart is a poor aerofoil, but a good enough one if the wings are even and the nose is not too heavy.

Weight is the paper. A slightly heavier nose keeps the dart from stalling. A paper-clip on the tip is a trim tab.

Thrust is your throw. After that there is only glide. Drag is everything that is not smooth: a puffy crease, a bent wing, a throw that starts the plane tumbling.

If it dives, unfold a millimetre of up-elevator at the back of each wing. If it stalls, add a little weight to the nose. If it rolls, the wings are not the same. Make them the same.



side view • even wings, a little weight in the nose